

up on both terraces and check the transmission rate with my laptop... I had to stand at the edge of tall buildings, and could have fallen 40 metres with even the slightest of pushes. Planting that antenna was risky."

Thankfully, we were provided with a makeshift ladder when we went there to check it out!

The Prize

After a few bouts of trial and error, Ravi managed to align the antennas and ping each machine from the other. It was then time for jubilation: he finally saw his dad's EAI (Enterprise Application Integration) solution working from both machines. This meant someone sitting in the administrator's office could easily check and operate on inventory maintenance in the godown. PHARMA ASSIST, the EAI solution Ravi's father uses, comes with multi-user support across a network.

Ravi has managed to set up videoconferencing and has enabled sharing of inventory data between the two machines with a throughput of about 80 Mbps. This is commendable, considering that the theoretical maximum of 802.11 Super G is 108 Mbps. Ravi admits, however, with a sigh of relief, that he has been lucky not to have interference from other Wi-Fi networks—something Wi-Fi (especially Super G) system administrators are plagued by.

Entrepreneurship, Anybody?

What started off as a hobby to help his dad's business has blossomed into a prospective business plan. Ravi is currently working on a plan to make higher-gain antennas to increase the propagation distance of Wi-Fi.

The road ahead for this is not easy. Making a project for his own use is one thing; product placement and marketing is a different ballgame altogether. For this, Ravi has teamed up with XITECH Solutions, a company that provides networking solutions, to help him explore business opportunities. Work continues on the tech front in his own testing centre in a Bangalore suburb.

Ravi's ideas have many takers, including a Government of Karnataka project for which he made a presentation of his ideas. The project—called GramPatra—aims to give villagers land record information using Wi-Fi, as part of a larger e-governance plan.

There is a broad smile on Ravi's face when he says he is working on finding loopholes in the current system at GramPatra and developing high-gain directional and omni-directional antennas for the project.

Of late, he has been approached by network solution providers asking if he could work on some turnkey projects.

"My plans for the immediate future are to

Jargon Buster

Antenna Gain: The ratio of the signal, usually expressed in dB, received or transmitted by the antenna as compared to an isotropic antenna, when both are supplied the same power. An isotropic antenna is a theoretical antenna: if it existed, it would radiate a signal equally in all directions.

Directional Antenna: An antenna that radiates waves in only one direction, such that energy is not transmitted in unwanted directions.

Lossy Cable: A cable or wire in which a signal experiences

considerable "loss" while being transmitted, meaning the signal strength is reduced at the end point of the cable.

Omni-Directional Antenna: An antenna that transmits signals equally in all directions.

RJ58: An inexpensive, coaxial TV cable suitable for cable TV transmission across short distances.

Super G Wi-Fi: An extension of the 802.11 g Wi-Fi standard, which transmits in the 2.4 GHz band and has a theoretical data transfer rate of 108 Mbps.

take Wi-Fi installations to apartments, societies, hotels, and the corporate world. This would be on the existing technology, and I would work on the R&D of my topology and come up with solutions offering longer ranges, especially for specific corporate needs," says Ravi.

The fascination with the building of one's own antenna seems universal. Network geeks in the US have come up with what they call a "cantenna" for Wi-Fi. The "C" simply stands for "can." A can of anything, but most popularly, an empty Pringles chips can! This, however, is another story for another time.

For Ravi, experimenting is a not a one-off thing. He had earlier designed a pollution-reducing add-on to two wheelers using calcium carbonate, which converts carbon monoxide to the less harmful carbon dioxide. That project had won him awards.

Ravi comes across as an extremely unassuming person—it was only when his lecturer approached a news daily that he got the media attention his innovation deserved. Asked where he gets his ideas from, given his commerce background, he says "common sense," with no second thoughts and not so much as a blink of an eye. It's a pity this breed of sense isn't more pervasive!

We wish Ravi all the best, and hope to see a lot more of the DIY spirit! ☑

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The Costs

Two WLAN cards	Rs 4,000
One access point	Rs 9,500
Connecting wire for 30 m	Rs 300
Utensils used as reflectors	FREE!
Total: (Less than)	Rs 15,000